

5	(a)		Current = Power / pd = $3 \times 10^6 / 125 \times 10^3 = 24 \text{ A}$	1
			Power lost = $I^2 R = (24)^2 (80) = 4.6 \times 10^4 \text{ W}$	1
	(b)		Energy lost = (46 kW) (24 h) = 1104 kWh	1
			Money lost = (1104) (\$0.25) = \$276	1
	(c)		No power lost in transformer due to eddy currents in the core/windings	1
			No leakage of magnetic flux	1
6(a)		• Electrons in the gas atoms interact with the photons and the photon energy		